

★ WEEK 4 · PART 3 · PRACTICE

ARIMA in Practice, Worked by Hand

Select the Order, Then Prove It Out of Sample

AIC and BIC, RMSE against a naive benchmark, and coverage

Part 3's full routine, with the arithmetic shown. We score candidate ARIMA orders with AIC and BIC, then evaluate the chosen model out of sample - RMSE against a naive forecast, interval coverage, and the crucial distinction between modeling prices and returns.

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PART 3 · APPLICATIONS

Fit, Then Prove It Holds Up

Case A: score candidate ARIMA orders with AIC and BIC, and see them trade fit against parsimony. Case B: evaluate out of sample - RMSE versus a naive forecast, prediction-interval coverage, and why prices and returns need different scrutiny. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Selection balances fit and complexity

AIC and BIC both reward likelihood and punish parameters - BIC more harshly. The lowest score wins, and they can disagree.

2

Only out-of-sample results count

In-sample fit always improves with parameters. The honest test is forecast error on data the model never saw.

3

Prices and returns need different scrutiny

On price levels an ARIMA forecast collapses to the naive one. The real edge, if any, shows up on returns - out of sample.

Everything here is a calculator exercise: one AIC/BIC table, one out-of-sample RMSE, one coverage count.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from candidate models or a test set.
- Score them, or measure forecast error.
- Then decide what to trust.

What you should watch

- How AIC and BIC penalize parameters differently.
- How RMSE compares to a naive benchmark.
- Why prices flatter a model and returns test it.

Selecting the Order

Fit against parsimony

With several plausible ARIMA orders, we need a principled choice. AIC and BIC both start from the model's likelihood and subtract a penalty for each parameter - BIC's penalty grows with the sample size, so it leans toward simpler models. We score four candidates and watch the two criteria nick differently

The candidates

- From a random walk to ARIMA(2,1,1).
- Each with its log-likelihood.
- Two hundred observations.

What we'll do by hand

- Compute AIC for each candidate.
- Compute BIC for each.
- See which order each criterion picks.

AIC and BIC

Reward fit, punish parameters

$$\text{AIC} = -2 \ln L + 2k, \quad \text{BIC} = -2 \ln L + k \ln T \quad (\text{fit } vs \text{ parameter count } k)$$

The shared part

- Both start from minus-two log-likelihood.
- Lower means a better fit.
- Both add a penalty for parameters.

The difference

- AIC's penalty is a flat two per parameter.
- BIC's is the log of the sample size.
- So BIC punishes complexity harder.

Score the Candidates

Four orders, two criteria

order	logL	k	AIC	BIC
(0,1,0)	-305	1	612.0	615.3
(1,1,0)	-298	2	600.0	606.6
(1,1,1)	-296	3	598.0	607.9
(2,1,1)	-295.5	4	599.0	612.2

Each AIC is minus-two times the log-likelihood plus twice the parameter count; each BIC swaps in the log of 200, about 5.3, per parameter. Reading down the last two columns reveals where each criterion bottoms out.

Where They Disagree

AIC richer, BIC simpler

AIC $\rightarrow (1, 1, 1)$ (598.0), BIC $\rightarrow (1, 1, 0)$ (606.6) (BIC's $\ln T$ penalty prefers parsimony)

AIC's pick

- Lowest at ARIMA(1,1,1), 598.0.
- It tolerates the extra MA term.
- Because its penalty is mild.

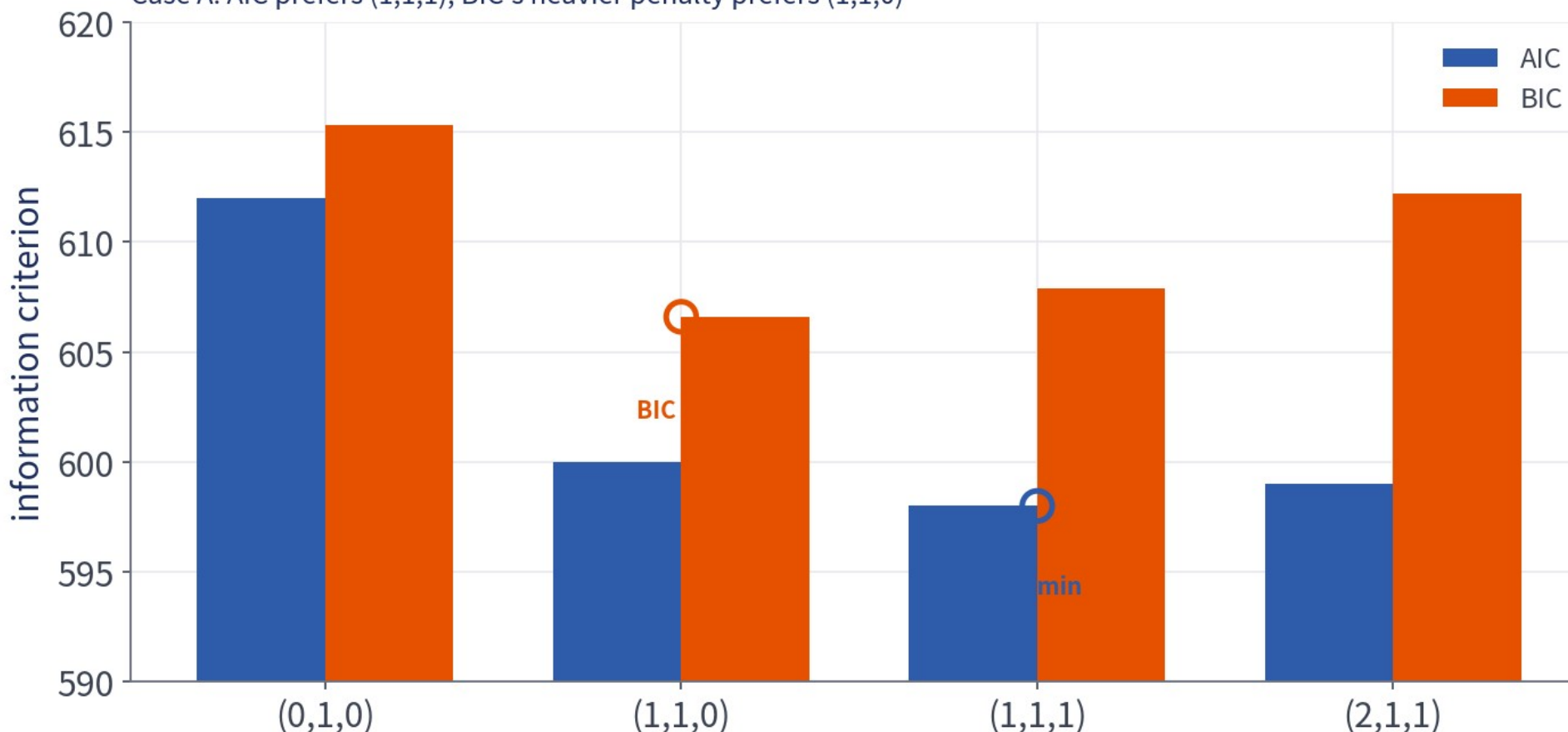
BIC's pick

- Lowest at ARIMA(1,1,0), 606.6.
- The extra term isn't worth the penalty.
- So it chooses the simpler model.

Two Criteria, Two Choices

AIC (blue) bottoms out at ARIMA(1,1,1), rewarding the slightly richer fit. BIC (orange), with its heavier per-parameter penalty, bottoms out one model earlier at (1,1,0). Both rise again for the over-parameterized (2,1,1). When they disagree, the simpler BIC choice usually forecasts more reliably.

Case A: AIC prefers (1,1,1); BIC's heavier penalty prefers (1,1,0)



Manual or Automated

Both land in the same place

Automated search

- Tools like `auto_arima` grid over orders.
- They minimize AIC or BIC for you.
- Fast, and a sensible default.

Manual judgment

- Reading ACF and PACF suggests orders too.
- Prefer the simpler model when scores are close.
- Automation is a starting point, not the last word.

Proving It Out of Sample

The only test that counts

A model that fits the past can still forecast badly. So we hold out a test set and measure the error there - root mean squared error - against the simplest possible benchmark, the naive forecast that just repeats the last value. We compute both, check whether the prediction interval really covers, and confront the prices-versus-

The evaluation

- Five held-out test points.
- ARIMA errors versus naive errors.
- Then count interval coverage.

What we'll do by hand

- Compute RMSE for both.
- Compare the two.
- Check the 95% band and the prices caveat.

Out-of-Sample RMSE

Error on data the model never saw

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y_t - \hat{y}_t)^2} \quad (\text{on the held-out test set, never the training data})$$

The rule

- Train on the first part, test on the rest.
- Never score on the training data.
- Rolling-origin repeats this many times.

The benchmark

- Compare to the naive last-value forecast.
- Beating it is the minimum bar.
- Many models quietly fail to.

ARIMA vs Naive

The out-of-sample scoreboard

test point	1	2	3	4	5
ARIMA error	+0.3	-0.5	+0.2	+0.4	-0.3
naive error	+0.5	-0.7	+0.4	+0.6	-0.5

$$\text{ARIMA } \sqrt{0.126} = \mathbf{0.355} < \text{naive } \sqrt{0.302} = \mathbf{0.550} \quad (\approx 35\% \text{ better out-of-sample})$$

Squaring, averaging, and rooting the errors gives an ARIMA RMSE of 0.355 against the naive 0.550 - about a 35% improvement out of sample. The model earns its keep on this return series.

Coverage, and Prices vs Returns

Two final checks

on price levels: $\hat{y}_{T+1} \approx y_T$ (the naive forecast) $\Rightarrow$ no edge; test on RETURNS

Interval coverage

- The 95% band was plus-or-minus 0.7.
- All five test errors fell inside.
- So the interval is roughly calibrated.

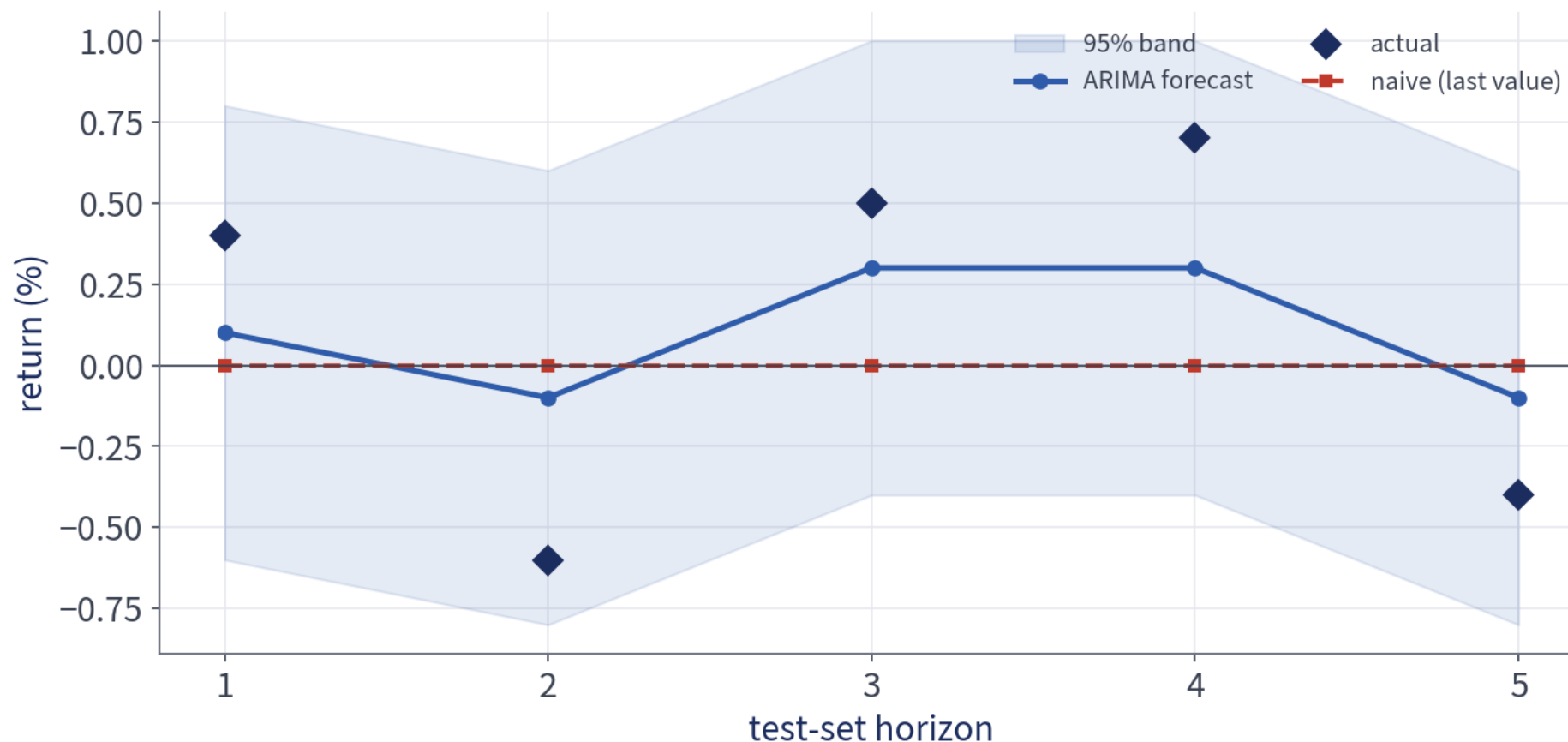
The prices caveat

- On price levels, ARIMA just echoes the last price.
- That's the naive forecast - no edge.
- The honest test is RMSE on returns.

Tracking the Test Set

The blue ARIMA forecast, with its 95% band, follows the held-out actuals (diamonds) more closely than the flat naive forecast (dashed red). Every actual falls inside the band - coverage holds - and the tighter tracking is what turns into the lower RMSE. This is the out-of-sample proof.

Case B: ARIMA tracks the test actuals better than the naive forecast



The Practical Distinction

Prices flatter, returns test

Why prices mislead

- A price is a random walk plus drift.
- The best forecast is nearly the last value.
- In-sample fit looks great but adds nothing.

Why returns are honest

- Any genuine structure lives in the returns.
- Out-of-sample RMSE reveals it - or its absence.
- Always benchmark against the naive forecast.

The ARIMA Pipeline

Weeks 2 to 4 in our course

The full routine is compact. Transform to stabilize the variance and clean the data. Difference to stationarity and identify the orders, letting AIC or BIC decide. Fit, then check residuals and score out of sample against a naive benchmark, verifying interval coverage. And always ask whether you're modeling a price or a return - the answer

The selection (Case A)

- AIC picks (1,1,1), BIC picks (1,1,0).
- Prefer the simpler when close.
- Automation as a starting point.

The evaluation (Case B)

- ARIMA RMSE 0.355 beats naive 0.550.
- Coverage holds: 5 of 5 inside.
- Test on returns, not prices.

Reproduce the Selection and the Evaluation

Direct Claude Code to check your arithmetic

GOAL Score candidates by AIC/BIC, then evaluate out of sample against a naive forecast.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - selection:  
  * with T=200 and logL for orders (0,1,0),(1,1,0),  
    (1,1,1),(2,1,1), print each AIC and BIC;  
  * report which order each criterion picks.  
CASE B - evaluation:  
  * for ARIMA errors [0.3,-0.5,0.2,0.4,-0.3] and naive  
    [0.5,-0.7,0.4,0.6,-0.5], print both RMSEs;  
  * count how many errors fall in a +/-0.7 band.  
Then plot AIC/BIC and the forecast vs actuals.
```

Check against the slides

- AIC min (1,1,1)=598.
- BIC min (1,1,0)=606.6.
- RMSE 0.355 vs 0.550.
- coverage 5/5.

Deliverable: part3_practice.py whose printed numbers match the AIC/BIC table and the RMSEs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do AIC and BIC differ?

BIC's per-parameter penalty grows with the sample size

Q2 What's the honest test of a forecast?

Out-of-sample error versus a naive benchmark

Q3 Why test on returns, not prices?

On prices, ARIMA collapses to the naive forecast

All three come from the AIC/BIC selection, the out-of-sample RMSE, and the prices-versus-returns caveat.

Three Pitfalls to Avoid

Where ARIMA practice trips people

Common mistakes

- "Pick the model with the best in-sample fit."
- "A low AIC guarantees good forecasts."
- "Great fit on prices means a good model."

The reality

- In-sample fit always improves - score out of sample.
- Beat the naive benchmark, or you have nothing.
- Prices flatter any model; test on returns.

Two Calculations to Remember

Part 3 applications in one screen

1

Score the orders, mind the penalty

Across four candidates, AIC bottomed at ARIMA(1,1,1) at 598 while BIC, with its log-of-T penalty, bottomed one model earlier at (1,1,0). When the two criteria disagree, the simpler BIC choice usually forecasts more reliably.

2

Prove it out of sample

On five held-out points, the ARIMA RMSE of 0.355 beat the naive forecast's 0.550 - about 35% better - and all five errors fell inside the 95% band, so coverage held. That out-of-sample win, not the in-sample fit, is the real evidence.

3

Always ask: price or return?

On price levels an ARIMA forecast just echoes the last value - the naive forecast - so any apparent skill is illusory. The honest evaluation is out-of-sample RMSE on returns, benchmarked against naive.

Week 4 complete — the full ARIMA and SARIMA pipeline, selected by information criteria and proven out of sample.